

Assignment 4

Josh Suber
RIN: 662038070

CSCI 4100

Contents

1	Exercise 2.4	3
1.1	(a) Show that $d_{VC} \geq d + 1$	3
1.2	(b) Show that $d_{VC} \leq d + 1$	3
1.3	Conclusion	4
2	Problem 2.3	4
2.1	(a) Positive or Negative Ray	4
2.2	(b) Positive or Negative Interval	5
2.3	(c) Two Concentric Spheres in $\mathbb{R}^d$	5
2.4	Conclusion	5
3	Problem 2.8	6
3.1	Key Concept: Growth Function and VC Dimension	6
3.2	Analysis of Each Function	6
3.3	Conclusion	7
4	Problem 2.10	7
4.1	Step 1: Showing $m_{\mathcal{H}}(2N) \leq m_{\mathcal{H}}(N)^2$	7
4.2	Step 2: Generalization Bound	8
4.3	Conclusion	8
5	Problem 2.12	8
5.1	Solution	8
5.2	Step 1: Set the generalization error bound and confidence level	9
5.3	Step 2: Apply the generalization bound for large N	9
5.4	Step 3: Solve for N	9
5.5	Conclusion	10

1 Exercise 2.4

1.1 (a) Show that $d_{VC} \geq d + 1$

To show that $d_{VC} \geq d + 1$, we need to find $d + 1$ points that can be shattered by the perceptron.

We are asked to construct a set of $d + 1$ points in $\mathbb{R}^d$ that can be shattered. The hint suggests we think about this in terms of linear independence. The $d + 1$ points must form a nonsingular matrix, meaning that they are linearly independent.

Approach:

- Consider $d + 1$ points in $\mathbb{R}^d$ and augment them with the constant coordinate $x_0 = 1$. This gives us $d + 1$ points in $\mathbb{R}^{d+1}$.
- These points can be arranged into a $(d + 1) \times (d + 1)$ matrix. If this matrix is nonsingular, the perceptron can shatter the points, because the points are linearly independent.
- Because the points are independent, we can apply all 2^{d+1} dichotomies and the perceptron will be able to correctly classify all possible labelings.

Thus, $d + 1$ points can be shattered, and we conclude that $d_{VC} \geq d + 1$.

1.2 (b) Show that $d_{VC} \leq d + 1$

Now, we need to show that no set of $d + 2$ points in $\mathcal{X}$ can be shattered by the perceptron.

The idea is that if we have $d + 2$ points in $\mathbb{R}^{d+1}$, these points are guaranteed to be linearly dependent. This follows from the fact that we have $d + 2$ vectors in a $(d + 1)$ -dimensional space, which means at least one of the points can be written as a linear combination of the others.

Process:

- Suppose we have $d + 2$ points in $\mathcal{X}$. These points are vectors in $\mathbb{R}^{d+1}$.

- Since they are linearly dependent, one of the points can be expressed as a linear combination of the others.
- The classification of this dependent point will be determined by the classification of the other points, so not all dichotomies are possible. In particular, there will always be at least one dichotomy that cannot be implemented by the perceptron.
- Therefore, the perceptron cannot shatter $d + 2$ points.

Thus, we have shown that $d_{VC} \leq d + 1$.

1.3 Conclusion

Combining parts (a) and (b), we conclude that the VC dimension of the perceptron is exactly $d + 1$. The perceptron can shatter $d + 1$ points, but cannot shatter $d + 2$ points.

2 Problem 2.3

2.1 (a) Positive or Negative Ray

Model: $\mathcal{H}$ contains functions that are +1 on $[a, \infty)$ (for some a), together with functions that are +1 on $(-\infty, a]$ (for some a).

Maximum Number of Dichotomies, $m_{\mathcal{H}}(N)$: In this case, we have a one-dimensional problem where each point can be classified as either +1 or -1 depending on whether it is to the left or right of the threshold a . For N points, the number of dichotomies is $N + 1$, as we can place the threshold a before the first point, between any two points, or after the last point, yielding $N + 1$ dichotomies.

VC Dimension, d_{VC} : The VC dimension is the largest number of points that can be shattered. Since the number of dichotomies is $N + 1$, the VC dimension is 1. For $N = 1$, two dichotomies can be implemented, and for $N = 2$, not all possible dichotomies can be implemented. Therefore, $d_{VC} = 1$.

2.2 (b) Positive or Negative Interval

Model: $\mathcal{H}$ contains functions that are +1 on an interval $[a, b]$ and -1 elsewhere, or -1 on an interval $[a, b]$ and $+1$ elsewhere.

Maximum Number of Dichotomies, $m_{\mathcal{H}}(N)$: In this case, each dichotomy is defined by selecting two points a and b to define the interval. The points within the interval are classified as $+1$, and those outside are classified as -1 . For N points, we can pick a before any point and b after any point, leading to $\binom{N+1}{2}$ possible intervals. The number of dichotomies grows quadratically with N , specifically $m_{\mathcal{H}}(N) = \binom{N+1}{2} + 1$.

VC Dimension, d_{VC} : The VC dimension in this case is 2. For $N = 2$, the points can be shattered by choosing an appropriate interval. For $N = 3$, it is not possible to implement all eight dichotomies, so $d_{VC} = 2$.

2.3 (c) Two Concentric Spheres in $\mathbb{R}^d$

Model: $\mathcal{H}$ contains functions that are $+1$ inside the region defined by $a \leq \sqrt{x_1^2 + \cdots + x_d^2} \leq b$ (inside two concentric spheres) and -1 elsewhere.

Maximum Number of Dichotomies, $m_{\mathcal{H}}(N)$: In this case, the decision boundary is based on two spheres. The points between the two spheres are classified as $+1$, and those outside as -1 . For each pair of spheres, the number of dichotomies depends on the relative positioning of the two spheres. While many dichotomies are possible for small N , the number of dichotomies is still limited by the geometry of the model.

VC Dimension, d_{VC} : For d -dimensional spheres, the VC dimension is $d+1$. For $d+1$ points, it is possible to shatter the set by adjusting the radii of the two spheres. For more than $d+1$ points, not all dichotomies can be implemented because of the linear dependencies between the points in $\mathbb{R}^d$. Therefore, $d_{VC} = d+1$.

2.4 Conclusion

- (a) The VC dimension of the positive or negative ray model is $d_{VC} = 1$.

- (b) The VC dimension of the positive or negative interval model is $d_{VC} = 2$.
- (c) The VC dimension of the two concentric spheres model in $\mathbb{R}^d$ is $d_{VC} = d + 1$.

3 Problem 2.8

3.1 Key Concept: Growth Function and VC Dimension

A valid growth function $m_{\mathcal{H}}(N)$ counts the maximum number of dichotomies that a hypothesis set $\mathcal{H}$ can implement on N points. For $N \leq d_{VC}$, the growth function equals 2^N , since all dichotomies can be implemented. For $N > d_{VC}$, the growth function must be polynomially bounded by N^d , where d is the VC dimension.

3.2 Analysis of Each Function

- $1 + N$: This function grows linearly with N . A valid growth function must grow exponentially for small N (up to the VC dimension), so this function cannot be a valid growth function. It underestimates the number of possible dichotomies.
- $1 + N + \frac{N(N-1)}{2}$: This function is quadratic in N . Such a function is plausible as a growth function for a hypothesis set with VC dimension 2. It is consistent with the expected polynomial growth for $N > d_{VC}$. This could be a valid growth function.
- 2^N : This is the maximum possible number of dichotomies for any hypothesis set, meaning it represents the case where the VC dimension is infinite or for small N , where $N \leq d_{VC}$. This is a valid growth function.
- $2 \lfloor \sqrt{N} \rfloor$: This function grows sublinearly as $\sqrt{N}$. Since a valid growth function should grow exponentially for small N , this function cannot represent a valid growth function.
- $2 \lfloor N^{1/2} \rfloor$: This function is similar to the previous one, growing sublinearly with N . This cannot represent a valid growth function.

- $1 + N + \frac{N(N-1)(N-2)}{6}$: This function is cubic in N . A cubic growth function is valid for a hypothesis set with VC dimension 3, as for large N , the growth function must be bounded by N^d . This could be a valid growth function.

3.3 Conclusion

- Valid growth functions:
 - $1 + N + \frac{N(N-1)}{2}$
 - 2^N
 - $1 + N + \frac{N(N-1)(N-2)}{6}$
- Invalid growth functions:
 - $1 + N$
 - $2 \lfloor \sqrt{N} \rfloor$
 - $2 \lfloor N^{1/2} \rfloor$

4 Problem 2.10

4.1 Step 1: Showing $m_{\mathcal{H}}(2N) \leq m_{\mathcal{H}}(N)^2$

Consider a set of $2N$ points, which we can split into two subsets of N points each. Call these subsets S_1 and S_2 . For the N points in S_1 , there are $m_{\mathcal{H}}(N)$ possible dichotomies (since $m_{\mathcal{H}}(N)$ is the number of dichotomies for N points).

Similarly, for the N points in S_2 , there are also $m_{\mathcal{H}}(N)$ possible dichotomies.

To classify the entire set of $2N$ points, we can independently choose a dichotomy for S_1 and a dichotomy for S_2 . The total number of ways to dichotomize the $2N$ points is the product of the number of dichotomies for S_1 and S_2 .

Thus, the total number of dichotomies for $2N$ points is at most:

$$m_{\mathcal{H}}(2N) \leq m_{\mathcal{H}}(N) \times m_{\mathcal{H}}(N) = m_{\mathcal{H}}(N)^2$$

This proves the required inequality.

4.2 Step 2: Generalization Bound

Next, we can derive a generalization bound that only involves $m_{\mathcal{H}}(N)$. By using Sauer's Lemma and recognizing that the number of hypotheses grows polynomially with N after reaching the VC dimension threshold, we can relate the generalization error to the complexity of the hypothesis set $\mathcal{H}$, as measured by $m_{\mathcal{H}}(N)$.

For a sample of size N , the probability of error outside the training set (i.e., the generalization error) is influenced by $m_{\mathcal{H}}(N)$:

$$P(\text{bad generalization}) \leq \frac{m_{\mathcal{H}}(N)}{N}$$

This inequality provides a generalization bound that shows the error decreases as the number of training points N increases, and it also depends on the complexity of the hypothesis set, as reflected in $m_{\mathcal{H}}(N)$.

4.3 Conclusion

We have shown that $m_{\mathcal{H}}(2N) \leq m_{\mathcal{H}}(N)^2$, and from this, we derived a generalization bound that only involves $m_{\mathcal{H}}(N)$.

5 Problem 2.12

5.1 Solution

We will use the generalization bound to determine the required sample size N . The generalization bound is given by:

$$\text{generalization error} \leq \hat{E} + \sqrt{\frac{8}{N} \ln \frac{4m_{\mathcal{H}}(2N)}{\delta}}$$

where:

- $\hat{E}$ is the in-sample error,
- N is the sample size,

- δ is the confidence parameter,
- $m_{\mathcal{H}}(N)$ is the growth function, and
- d_{VC} is the VC dimension.

5.2 Step 1: Set the generalization error bound and confidence level

We are given that the generalization error should be at most 0.05 with a confidence level of 95%. This means:

- The generalization error $\epsilon = 0.05$,
- The confidence $\delta = 0.05$ (since 95% confidence corresponds to $1 - \delta = 0.95$).

5.3 Step 2: Apply the generalization bound for large N

For large N , the growth function $m_{\mathcal{H}}(N)$ behaves as $m_{\mathcal{H}}(N) \leq N^{d_{VC}}$, and the generalization bound simplifies to:

$$\epsilon \leq \sqrt{\frac{8}{N} \ln \left(\frac{4N^{d_{VC}}}{\delta} \right)}$$

Substitute $d_{VC} = 10$, $\epsilon = 0.05$, and $\delta = 0.05$:

$$0.05 \leq \sqrt{\frac{8}{N} \ln \left(\frac{4N^{10}}{0.05} \right)}$$

5.4 Step 3: Solve for N

Square both sides of the inequality to eliminate the square root:

$$0.0025 \leq \frac{8}{N} \ln \left(\frac{4N^{10}}{0.05} \right)$$

Multiply both sides by N :

$$0.0025N \leq 8 \ln \left(\frac{4N^{10}}{0.05} \right)$$

Now, solve this equation numerically or by approximation to find N . By approximation, we find that the required sample size N is around 4500.

5.5 Conclusion

To achieve a generalization error of at most 0.05 with 95% confidence, the required sample size is approximately $N = 4500$.